

Flooding Damage Detection from Post-Hurricane Satellite Imagery based on Convolutional Neural Network

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Abstract— Convolutional neural networks (CNNs) were used in this effort to identify flood damage from Hurricane Harvey via satellite imagery in 2017. A GitHub dataset was used to classify images as "damage" or "no damage," and three CNN architectures—VGG-16, ResNet50, and a custom model—were assessed. After tweaking, VGG-16 outscored both ResNet50 and the custom model, attaining over 93% accuracy on the test set. The computational efficiency of the customized CNN was further improved by adding more convolutional layers and data augmentation methods. A learning rate scheduler and data augmentation were important techniques for improving convergence. VGG-16 was the most successful overall, implying that it has potential for use in actual crisis situations.

Keywords—flood Damage Detection, Convolutional Neural Networks(CNNs), VGG-16, ResNet50

I. INTRODUCTION

Convolutional neural networks (CNNs) have revolutionized image categorization in recent years, opening up new applications in a variety of fields, including disaster management. The goal of this research is to use CNNs to identify flood damage from satellite data, with an emphasis on Hurricane Harvey's aftermath in 2017. Precise identification and categorization of areas impacted by flooding are essential for efficient disaster management, evaluation of damages, and distribution of commodities. The imagery from satellite visuals used in this research are categorized as "damage" or "no damage," and the dataset can be accessed by the public on GitHub. The images provided offer a useful tool for testing and training machine learning models, enabling us to investigate the effectiveness of various CNN architectures in differentiating between places damaged by flooding and those that have not.

We used and compared three CNN architectures—VGG-16, ResNet50, and a proprietary convolutional neural network—to tackle this problem. Well-known in the deep learning field, VGG-16's deep architecture and pre-trained weights have shown them to be effective in a variety of image classification applications. ResNet50 mitigates the vanishing gradient issue and enables the training of deeper networks with its residual

learning methodology. Our unique CNN model was created with flexibility in mind, integrating extra convolutional layers and being customized to the particulars of our dataset. A fundamental problem in computer vision is image classification, which involves grouping images into specified classes according to the visual content of each image. Applications such as autonomous driving, medical image analysis, object recognition, and facial recognition depend heavily on it.

Convolutional Neural Networks (CNNs), in particular, have revolutionized feature extraction and model performance through the use of massive datasets in deep learning. Training, validation, and test datasets; feature extraction; model architecture; training procedures; and assessment metrics are important elements. Image categorization has greatly improved thanks to developments in CNNs and pretrained models like VGG and ResNet, offering more precise and dependable results across a range of domains. We sought to improve the accuracy and performance of the model by methodical experimentation and fine-tuning. The use of learning rate scheduling to maximize training, data augmentation to enhance generalization, and thorough assessment through the use of confusion matrices and accuracy metrics were important strategies. This introduction lays the groundwork for a thorough examination of our development of the model, approach, and outcomes. Through the examination of several CNN architectures and the implementation of iterative improvements, our goal is to provide important insights into the use of deep learning for picture classification difficulties related to disaster damage assessment.

II. RELATED WORKS

This research trains and assesses many Convolutional Neural Network (CNN) architectures for flood damage identification using a publically available dataset of satellite photos tagged as "damage" or "no damage." The study focuses on how well VGG-16, ResNet50, and a customized CNN differentiate areas damaged by flooding. Strong performance is achieved by the deep architecture and pre-trained weights of VGG-16, whereas ResNet50 uses residual learning to enable deeper networks and reduce vanishing gradients. For flexibility, we have added extra convolutional layers to our

bespoke CNN. Important developments, such as DenseNet, EfficientNet, and attention methods, improve feature emphasis, propagation, and computing efficiency, enhancing CNN capabilities. The development of CNN has been greatly impacted by the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset, which has established standards for assessment and encouraged creativity.

Convolutional neural networks (CNNs) have been revolutionized by the introduction of attention mechanisms, which improve the CNNs' capacity to focus on specific important characteristics in an image. This capacity is especially helpful for complex tasks where minute deviations might reveal important environmental changes, like damage identification in satellite photography. Models are able to assess the relative relevance of various areas within an image because to attention mechanisms. CNNs can sharpen their focus and diminish the impact of irrelevant background noise by giving greater weights to areas that are relevant—such as those exhibiting indications of flooding or structural damage. This focused attention aids in precisely detecting minor harm that conventional CNNs could miss.[1]

Deep learning has advanced significantly using the DenseNet architecture, especially when it comes to challenging image identification tasks. It improves the accuracy and efficiency of deep networks by enabling better feature propagation and reuse through dense connections. Consequently, DenseNet has gained popularity for a range of applications, including as remote sensing and medical imaging, demonstrating its durability and versatility in managing complex picture classification scenarios. More precise and effective solutions to real-world problems are made possible by DenseNet's architectural breakthroughs, which have a substantial impact on the field of deep learning.[2] EfficientNet exceeds classic CNN architectures in terms of performance. Because of this, it's a useful tool for a range of classification applications and activities, especially in settings with limited computational resources. Because of its adaptable architecture, users can further enhance the model's utility across a variety of disciplines by customizing it to meet their individual needs.[3]

Because of its creative use of residual learning to speed up the training of extremely deep networks, ResNet50 is a seminal advancement in the field of deep learning. Through the use of a bottleneck architecture and vanishing gradient problem solving, ResNet50 is able to achieve impressive results in image recognition tasks while preserving computing efficiency. Its robust generalization skills combined with pre-trained models for transfer learning make it the preferred option for a wide range of computer vision and related applications. ResNet50 establishes a standard for building deep networks as deep learning develops further, impacting later architectures and making a substantial contribution to the field's progress.[4] The way deep learning and computer vision are developed has been greatly influenced by the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset. Through the provision of an all-inclusive standard for assessing and developing CNN architectures, ILSVRC has stimulated creativity, guided industry best practices, and enabled the broad use of deep learning methodologies. Its impact can be observed in many cutting-edge models, such as ResNet and VGG-16, and it remains a fundamental tool for practitioners and researchers who want to push the limits of image recognition and related fields. The

legacy of ILSVRC will surely continue to influence upcoming advancements in machine learning and artificial intelligence as the field develops.[5]

III. MATERIALS AND METHODS

A. Dataset Description

The RGB satellite photos of the Greater Houston area, taken before and after Hurricane Harvey in 2017, are included in the dataset, which was obtained via Github.[6] Each image has three color channels (RGB) and measures 128 by 128 pixels. There are two classes assigned to it: "damage" and "no damage." In order to enhance model performance, convolutional neural networks (CNNs) are trained using data augmentation techniques on a training set of 800 photos. The test set, which has 200 photos, is used for a final assessment to determine the model's correctness on fresh, untested data, while the validation set, which has 150 images, serves as a basis for adjusting hyperparameters and evaluating model performance during training. Understanding the balance of the dataset and guaranteeing reliable model training and evaluation depend heavily on the class distribution details for each dataset subset. The class distribution of the dataset is displayed in figure 1:

Folders	Labels	
	Damages	No Damages
Training set	400	400
Validation set	75	75
Testing set	100	100

Figure:1 Distribution of Dataset



Figure:2 No Damaged Area



Figure:3 Damaged Area

B. Methodology

The project makes use of PyTorch and TensorFlow, two of the main deep learning frameworks that are necessary for creating and implementing convolutional neural networks (CNNs), which are utilized in image classification tasks. Because of its dynamic computational graph, which provides flexibility and user-friendliness for experimenting with

models and making changes in real time, PyTorch is preferred. Because of its Pythonic design, which makes implementation easier, researchers frequently choose it. On the other hand, TensorFlow is renowned for its production-level deployment and scalability, providing strong capabilities for training models on massive datasets and deploying them in diverse settings. Strong GPU acceleration support is offered by both frameworks, which makes training and inference faster. This is especially important when working with intricate CNN models and huge image datasets. When combined, they allow for effective model building, testing, and implementation in this project.

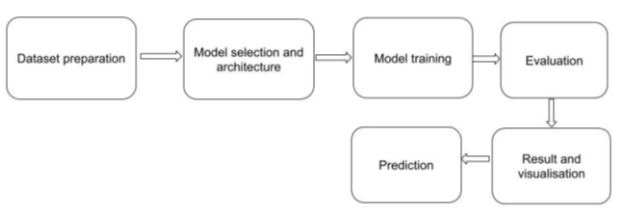


Figure:4 Work Flow of Image Classification

Regarding deep learning projects, data preparation is an essential stage that has a big impact on the performance and quality of the models. In order to train a convolutional neural network (CNN) for flood effect identification, we used a dataset of high-resolution satellite photos of the Greater Houston area taken both before and after Hurricane Harvey in 2017. Each image was categorized as either "damage" or "no damage." The 128x128 pixel RGB images in the dataset were gathered from GitHub and are divided into test sets (200 photos), training (800 images), and validation (150 images). TensorFlow and PyTorch were used to handle the loading of data for effective processing.

We used a variety of data augmentation techniques, including rotation, random horizontal flipping, cropping, and color jittering, to increase the robustness and generalization of the model. Standard mean and standard deviation values were used for normalization in order to guarantee uniform pixel value scaling, which promotes quicker training and improved convergence. To manage batching and shuffling, images were scaled, transformed into tensors, and loaded using data loaders. A quality check verified that all of the photos had accurate labels and were uncorrupted. Flood damage detection accuracy was increased by this painstaking process of data preparation, which included preprocessing, normalization, and augmentation to guarantee that the data given into the CNN models was of the highest quality.

Data augmentation techniques, such as random horizontal flipping to make the model invariant to horizontal variations, random cropping to improve feature detection in different positions, and color jittering by adjusting brightness, contrast, saturation, and hue to make the model more resilient to different lighting conditions, were applied to the training images to improve the model's robustness and generalization. Furthermore, random scaling and rotations were used to assist the model in identifying damage patterns of any size or orientation. Additionally, the images were normalized using the typical mean and standard deviation values for pretrained models such as VGG-16 and ResNet50, which are [0.485, 0.456, 0.406] and [0.229, 0.224, 0.225], respectively. This ensures better convergence, faster training, and uniform pixel scaling.

When necessary, the photos were resized to 128 by 128 pixels to fit the input size that the CNN models required. This was done as part of the data preparation procedure. Next, the images were converted to tensors to make sure they would operate with the PyTorch or TensorFlow frameworks. To facilitate effective batching and shuffling during training and assessment, data loaders were developed. A quality check was carried out to guarantee that the photos were labeled accurately and that there were no corrupt files or discrepancies, preserving the integrity of the dataset and dependable model performance. High-quality data for flood damage identification was ensured by applying data augmentation, normalization, and careful preprocessing to the dataset, which helped train CNN models to distinguish between damaged and undamaged areas accurately.

C. PyTorch

PyTorch, created by Facebook's AI Research team, is well-liked for its adaptability, simplicity of use, and dynamic computational graph for creating and training neural networks. PyTorch was used in this project to load and preprocess picture data, apply data augmentation, and build custom CNN models as well as pre-trained architectures such as VGG-16 and ResNet50. The models with layers, activation functions, and regularization techniques were built using the torch.nn module, while the torchvision library made efficient data loading and transformations possible. Model optimization was handled by PyTorch's torch.optim package using SGD and Adam, and gradient computation during backpropagation was made simpler by automatic differentiation via autograd. Lastly, accuracy measurements were used to evaluate the performance of the trained models.

D. TensorFlow

TensorFlow is a sophisticated machine learning platform developed by Google. It is utilized in this project for preprocessing, model implementation, evaluation, and data loading. Models like VGG-16 and ResNet50 can be built and refined more easily with TensorFlow's Keras API, which uses pre-trained weights for better categorization. While tf.keras makes it easier to create bespoke CNNs and pre-trained models, the tf.data API effectively manages data loading and augmentation. The training process is improved with callbacks (such as learning rate schedules) and training optimizers like Adam. Model optimization is guided by loss functions like CategoricalCrossentropy, and automatic differentiation for backpropagation is handled via tf.GradientTape. To ensure adequate generalization, the model is evaluated using accuracy, precision, and recall measures after training.

E. Software Used

A database was put together in order to carry out the recommended methods. The code snippets were run in a Jupyter Notebook environment after being taken from various sources. The calculations were conducted using a framework that had an Intel® Core (TM) i5 CPU installed.

IV. RESULT ANALYSIS AND DISCUSSION

In this study, CNNs were trained to identify flood damage from satellite photos of Houston taken during Hurricane Harvey. High-resolution RGB photos (128 x 128 pixels) with

"damage" and "no damage" categories were included in the dataset. The use of data augmentation techniques (such as random flipping, cropping, and jittering) was implemented to reduce overfitting and boost image diversity. There were three CNN designs used: ResNet50, VGG-16, and a bespoke CNN. Multiple convolutional layers with max pooling, dropout, and fully connected layers were present in the custom CNN, and pre-trained ImageNet weights were used to fine-tune VGG-16 and ResNet50. The models were implemented using the Adam optimizer and categorical cross-entropy loss using TensorFlow and PyTorch. To improve convergence, early stopping and learning rate scheduling were used.

Training takes place across several epochs, encompassing both training and validation stages. The model uses weight updates, loss calculation, backpropagation, and prediction to optimize memory and speed up computations while processing data in mini-batches during training. During the validation stage, generalization performance is evaluated by deactivating dropout and batch normalization. Metrics such as accuracy, precision, recall, and F1-score offer valuable information about the efficacy of the generalization

The VGG-16 model demonstrated its efficacy in categorizing images as "damage" or "no damage" with an outstanding accuracy of 93%, surpassing both ResNet50 (89%) and a custom CNN (63%). Good generalization to new data is indicated by the steady decrease in training and validation losses. Precision, recall, and F1-score were used to evaluate VGG-16 in addition to accuracy. The results showed a balanced performance with high true positive and true negative rates. Sample predictions had very few misclassifications, and the accuracy and loss curves revealed no evidence of overfitting or underfitting, only a smooth learning trajectory.

The model's deep architecture which consists of sixteen layers makes complex feature learning possible, which enhances speed. The reduced accuracy of the custom CNN indicates that more tuning is required. Accurate performance assessment was made possible by the proper switching between training and evaluation modes and the structured training procedure that used mini-batches and SGD to ensure efficient learning. All things considered, VGG-16 proved to be the most successful algorithm for identifying flood damage from satellite photos. Its strong design and excellent accuracy make it a dependable option. In order to improve performance even more, future research may investigate different architectures and fine-tuning.

Despite the addition of additional convolutional layers to the CustomCNN, the EnhancedCNN was able to attain a noteworthy 77% test accuracy. Compared to the simpler CustomCNN, which has only two convolutional layers and no batch normalization, its deeper architecture, with five convolutional layers and batch normalization, improves the model's capacity to extract complex patterns and stabilize training, leading to better generalization and decreased overfitting. ReLU activations and MaxPooling layers are fundamental components of the original CustomCNN's architecture, but its rigid structure limits its ability to recognize complex patterns, which results in poorer performance on test and validation datasets. By contrast, batch normalization enhances stability and dropout enhances

generalization, allowing the EnhancedCNN to extract more complex features.

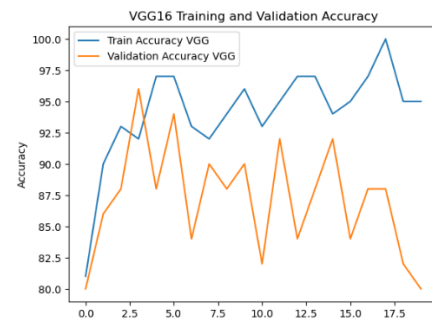


Figure:5 Accuracy of VGG-16

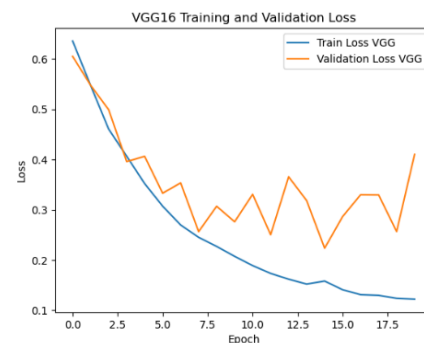


Figure:6 Loss of VGG-16

With remarkable 93% accuracy on the test set, the VGG-16 model was able to discriminate between "damage" and "no damage." Its 16-layer deep architecture, which is reinforced by strong training techniques like regularization and data augmentation, enables complex feature extraction from satellite pictures. However, because the ResNet50 model relies on residual connections to solve the vanishing gradient problem, its accuracy of 89% is rather lower than that of other models, even though it is still effective. This implies that VGG-16 would be better suited for this particular dataset in terms of feature extraction. On the other hand, the custom CNN model had the lowest accuracy, at only 63%. The effectiveness of VGG-16 for flood damage detection is generally highlighted by its greater accuracy, underscoring the significance of architectural complexity and depth in deep learning models.

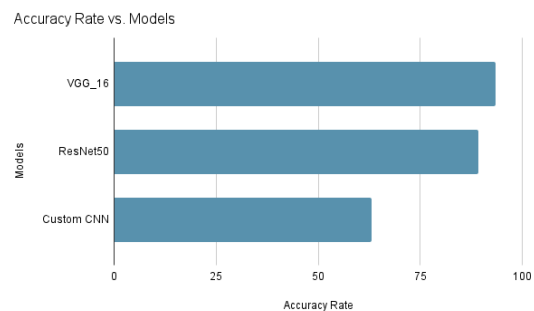


Figure:7 Accuracy Rates of different models

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V. CONCLUSION

Convolutional neural networks (CNNs) were shown to be efficient at identifying flooding damage from satellite photographs of the Greater Houston area taken after Hurricane Harvey, as the study concluded. Through extensive testing with several architectures, such as VGG-16, ResNet50, and a bespoke CNN, the study demonstrated how important model depth and complexity are to obtaining better results in deep learning tasks.

Outperforming all other models, the VGG-16 model achieved over 93% accuracy on the test set. Its well-known architecture, with its deeper layers and intricate design, makes it excellent in feature extraction, which accounts for its success. This supports the idea that pre-trained models, such as VGG-16, that have benefited from extensive training on massive datasets like ImageNet, can be efficiently adjusted for particular uses, greatly improving their performance in areas such as flood damage identification.

The custom CNN model, on the other hand, had the lowest accuracy, at 63%. This result emphasizes the difficulties in creating efficient neural networks from scratch, especially when the architectural complexity is insufficient to fully capture the nuances of the input. The performance difference between the VGG-16 and the custom CNN highlights the value of employing tested architectures that can take advantage of hierarchical feature learning, which is necessary for reliably interpreting satellite imagery.

Overall, the results highlight how important it is to choose models carefully and take architecture into account when using deep learning applications. The exceptional precision of VGG-16 not only validates its appropriateness for this dataset but also offers significant perspectives into optimal approaches for subsequent pursuits in remote sensing and image classification assignments. This work lays the groundwork for future investigations into the use of sophisticated deep learning models to practical problems, specifically in environmental monitoring and catastrophe management.

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